
Presentation Transcript: Alchemy of Waste Shells and Fruit Peels

[0:00 - 1:15] Introduction & Motivation

(Stage Direction: Stand near the 'Motivation' section. Speak with an engaging and confident tone.)

"Good morning, respected judges and fellow researchers. Today, I am proud to present our research titled: 'Alchemy of Waste Shells and Fruit Peels: Enhancing Pectin Flocculation and Heavy Metal Removal through Calcium Ion Crosslinking Effects'. Our research began with a dual environmental observation. On one hand, we have massive quantities of discarded oyster shells polluting our coastlines. On the other hand, many of our local lakes and ponds are suffering from severe turbidity and heavy metal pollution. This led us to our core motivation: using one type of waste to solve another. By combining discarded shells and fruit peels, we aimed to create a low-cost, sustainable water treatment solution.

[1:15 - 2:30] Methodology & Material Preparation

(Stage Direction: Gesture towards the 'Method/Material' flowchart.)

"To achieve this, we needed to prepare our materials meticulously. First, we dried fruit peels at 30°C and ground them to extract the raw pectin material. Simultaneously, we subjected the oyster shells to high-temperature calcination, followed by repeated washing and filtration to create a refined shell powder.

We then conducted column filtration and adsorption experiments. We packed a column with a 1:1 ratio of our processed powder and pectin, rinsed it with distilled water, and introduced a 5000 ppm Copper Sulfate solution to test its adsorption capabilities.

[2:30 - 4:00] Data Interpretation: Selecting the Optimal Base Material

(Stage Direction: Move to the 'Result & Discussion' section. Point clearly to the top three line graphs.)

"Before testing our composite, we had to determine the best base material. We evaluated three options over a 150-minute duration.

Please look at the top left graph tracking Eggshell Membrane Powder. While the overall trend is downward, showing a rapid drop in electrical conductivity in the first 50 minutes, it plateaued. The middle graph shows intact eggshell membrane, where the decrease in copper ions was unfortunately only slight.

However, our breakthrough is in the third graph: Fine Oyster Shell Powder. Here, the overall trend is sharply downward, demonstrating a consistently high rate of decrease over the entire 150 minutes. Based on these conclusive results, we selected oyster shell powder as the primary material for our subsequent, extended composite experiments.

[4:00 - 5:30] Synergistic Effects: Pectin and Calcium Crosslinking

(Stage Direction: Point to the bar charts featuring the fruit illustrations.)

"Next, we tested the 'alchemy' of combining this oyster shell powder with different fruit pectins—specifically orange, apple, and banana.

Look at our first bar chart, measuring the absorbance of a highly turbid Kaolin-Copper Sulfate solution. Before treatment, the absorbance was incredibly high, sitting at a baseline of 2.275. After introducing our composite material, the absorbance plummeted. Orange dropped to 0.091, apple to 0.083, and banana pectin proved the most effective, dropping to just 0.037.

This massive drop proves that our composite material, formed from oyster shells and pectin, has a remarkably strong potential to reduce water turbidity and improve water clarity.

[5:30 - 6:30] Scientific Objectivity & Limitations

(Stage Direction: Adopt a slightly more serious, academic tone.)

"As young scientists, we must also be completely transparent about our limitations. Our second bar chart shows the removal of Copper Ions specifically. While the data shows a promising drop from an initial 0.089 down to 0.009 across the pectins, we must state that regarding the total removal of heavy metal ions, it has not yet been fully verified. This is due to limitations in our current measurement methods and complex matrix effects in the solutions. Acknowledging this sets a clear path for our future testing.

[6:30 - 7:00] Application & Conclusion

(Stage Direction: Step back to address the whole panel, gesturing to the SDGs at the bottom.)

"Ultimately, our vision for this research is highly practical. We aim to manufacture this composite adsorbent into a convenient, capsule-shaped format. It is low-cost and operates with a tremendously low environmental impact.

By successfully turning agricultural and aquaculture waste into a clean-water solution, our project directly advances UN Sustainable Development Goals 6: Clean Water and Sanitation, 11: Sustainable Cities and Communities, and 15: Life on Land.

Thank you for your time. I am now happy to answer your questions."

Q&A 速查表：廢棄貝殼與果皮的鍊金術 (Alchemy of Waste Shells and Fruit Peels)

1. 研究動機 (Motivation)

- **EN:** What inspired you to use oyster shells and fruit peels for this research?
- **CH:** 是什麼啟發您使用牡蠣殼與果皮來進行這項研究？
- **Answer (EN):** We observed large quantities of discarded oyster shells and turbid local lakes. Our motivation was to "use one type of waste to solve another."
- **Answer (CH):** 我們觀察到大量被丟棄的牡蠣殼以及混濁的湖泊池塘。我們的動機是「用一種廢棄物來解決另一種環境問題」。

2. 核心目標 (Core Aim)

- **EN:** What is the main objective of combining oyster shells and pectin?
- **CH:** 結合牡蠣殼與果膠的主要目標是什麼？
- **Answer (EN):** To enhance pectin flocculation and heavy metal removal through calcium ion crosslinking effects.
- **Answer (CH):** 透過鈣離子的交聯效應，來提升果膠的絮凝表現以及重金屬的移除效率。

3. 材料製備 - 果膠 (Material Preparation - Pectin)

- **EN:** How did you process the fruit peels to get your pectin raw material?
- **CH:** 您如何處理果皮以取得果膠原料？
- **Answer (EN):** We dried the fruit peels at 30°C until dry, then ground and sieved them.
- **Answer (CH):** 我們將果皮在 30°C 下烘乾，然後進行研磨與過篩。

4. 材料製備 - 牡蠣殼 (Material Preparation - Oyster Shells)

- **EN:** What steps were involved in preparing the oyster shell powder?
- **CH:** 製備牡蠣殼粉的步驟有哪些？
- **Answer (EN):** We used high-temperature calcination on the shells, followed by repeated washing and filtration with distilled water.
- **Answer (CH):** 我們對牡蠣殼進行高溫煅燒，接著用蒸餾水進行反覆的清洗與過濾。

5. 基礎材料選擇 (Base Material Selection)

- **EN:** Why did you choose fine oyster shell powder over eggshell membrane powder?
- **CH:** 為什麼您選擇細牡蠣殼粉而不是蛋殼膜粉？
- **Answer (EN):** Because the fine oyster shell powder showed a consistently high rate of decrease in electrical conductivity over 150 minutes.
- **Answer (CH):** 因為細牡蠣殼粉在 150 分鐘內，其電導率呈現出持續且高效率的下降趨勢。

6. 吸附實驗 (Adsorption Experiment)

- **EN:** What solution did you use to test the adsorption capacity in your columns?
- **CH:** 您在管柱實驗中使用什麼溶液來測試吸附能力？
- **Answer (EN):** We used a 5000 ppm Copper Sulfate solution.
- **Answer (CH):** 我們使用的是 5000 ppm 的硫酸銅溶液。

7. 水質改善指標 (Water Clarity Indicator)

- **EN:** How did you measure the improvement in water clarity?
- **CH:** 您如何測量水質清晰度的改善程度？
- **Answer (EN):** We measured the reduction in absorbance (Abs) of a Kaolin-Copper Sulfate solution.
- **Answer (CH):** 我們透過測量「高嶺土-硫酸銅溶液」的吸光度 (Abs) 降低程度來評估。

8. 最佳果膠表現 (Best Performing Pectin)

- **EN:** Among the different fruit pectins tested, which one performed the best?
- **CH:** 在測試的不同水果果膠中，哪一種表現最好？
- **Answer (EN):** Banana pectin performed the best, dropping the absorbance from 2.275 down to 0.037.
- **Answer (CH):** 香蕉果膠表現最好，將吸光度從 2.275 大幅降至 0.037。

9. 複合材料效果 (Composite Material Effect)

- **EN:** What is the main conclusion from your absorbance tests?
- **CH:** 關於您的吸光度測試，主要的結論是什麼？
- **Answer (EN):** The composite material has strong potential to reduce water turbidity and improve water clarity.
- **Answer (CH):** 該複合材料在降低水質混濁度與改善水質清晰度方面，展現了強大的潛力。

10. 銅離子移除數據 (Copper Ion Removal Data)

- **EN:** Did the composite material lower the presence of copper ions?
- **CH:** 複合材料是否有降低銅離子的存在？
- **Answer (EN):** Yes, the data showed a drop from around 0.089 to 0.009, but this requires further verification.
- **Answer (CH):** 是的，數據顯示從大約 0.089 降到了 0.009，但這還需要進一步驗證。

11. 研究限制 (Research Limitations / Scientific Objectivity)

- **EN:** Why do you state that the removal of heavy metal ions is not fully verified?
- **CH:** 為什麼您聲明重金屬離子的移除尚未被完全驗證？
- **Answer (EN):** Due to limitations in our current measurement methods and complex matrix effects in the solutions.
- **Answer (CH):** 這是由於我們目前測量方法的限制，以及溶液中複雜的基質效應 (Matrix effects) 所致。

12. 鈣離子的作用 (Role of Calcium Ions)

- **EN:** What is the specific chemical effect of the oyster shell in this composite?
- **CH:** 牡蠣殼在此複合材料中具體的化學作用是什麼？

- **Answer (EN):** The calcium ions from the shells provide crosslinking effects that significantly enhance the pectin's flocculation.
答案（英文）：貝殼中的鈣離子具有交聯作用，能顯著增強果膠的絮凝作用。
- **Answer (CH):** 牡蠣殼釋放的鈣離子提供了交聯效應 (Crosslinking effects)，能顯著提升果膠的絮凝作用。

13. 未來實際應用 (Practical Application)

- **EN:** What is your vision for the final product of this research?
- **CH:** 您對這項研究最終產品的願景是什麼？
- **Answer (EN):** We aim to create a capsule-shaped composite adsorbent that is low-cost and convenient to use.
- **Answer (CH):** 我們致力於創造一種膠囊狀的複合吸附劑，它具備低成本且使用方便的特性。

14. 環境價值 (Environmental Value)

- **EN:** What is the main environmental advantage of your composite adsorbent?
- **CH:** 您複合吸附劑的主要環境優勢是什麼？
- **Answer (EN):** It transforms agricultural and aquaculture waste into a solution, creating a very low environmental impact.
- **Answer (CH):** 它將農業與水產養殖廢棄物轉化為解決方案，對環境的影響非常低。

15. 聯合國永續發展目標 (SDGs Alignment)

- **EN:** Which UN Sustainable Development Goals does your project align with?
- **CH:** 您的專案符合哪些聯合國永續發展目標？
- **Answer (EN):** It aligns with SDG 6 (Clean Water and Sanitation), SDG 11 (Sustainable Cities), and SDG 15 (Life on Land).
- **Answer (CH):** 它符合 SDG 6 (淨水及衛生)、SDG 11 (永續城鄉) 以及 SDG 15 (保育陸域生態)。

💡 評審應對小建議 (Tips for Q&A) :

當評審問到第 11 題（關於重金屬移除未完全驗證）時，這其實是您的加分項！您可以自信地回答：*"As high school researchers, we want to be scientifically honest. The drop from 0.089 to 0.009 is promising, but we acknowledged 'matrix effects' (基質效應) to show we understand real-world water complexity."* 主動承認研究限制，會讓評審覺得您具備真正科學家的嚴謹態度與成熟度。祝您報告順利！